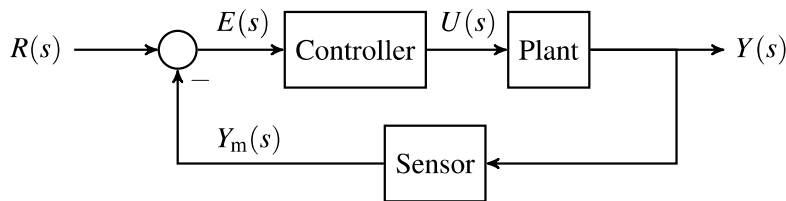


Chapter 3

Systems with Feedback

Ideally, we simplify a plant with block diagram algebra to a single transfer function, then design a control system with negative feedback such that the controlled block diagram is of the form



where we have three internal signals

1. $E(s)$ is the error signal

- When the feedback is negative the signal is the difference between the reference input and the measured output

$$E(s) = R(s) - Y_m(s)$$

2. $U(s)$ is the control effort

- The controller output that drives the plant

$$U(s) = \text{Controller} \cdot E(s)$$

3. $Y_m(s)$ is the measured output signal, which is used for feedback

$$Y_m(s) = \text{Sensor} \cdot Y(s)$$

- Often, the difference between the output signal and the measured signal is insignificant when designing a controlled process

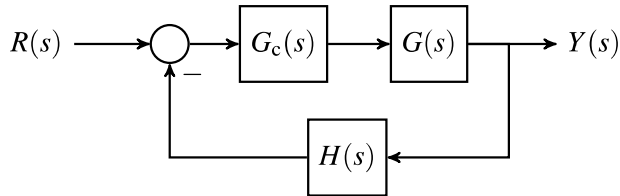
and two external signals

1. $R(s)$ is the reference input signal

2. $Y(s)$ is the controlled system output signal

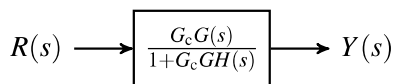
$$Y(s) = \text{Plant} \cdot U(s)$$

The above block diagram is accurate (but wordy) and is simplified to

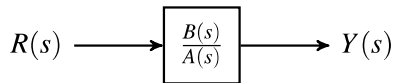


where the controller is $G_c(s)$, the plant is $G(s)$, the internal signals are the same, but not identified on the figure. The output feedback to the summation junction creates a loop in the system — this representation is a closed-loop model.

The diagram simplifies to a single block diagram



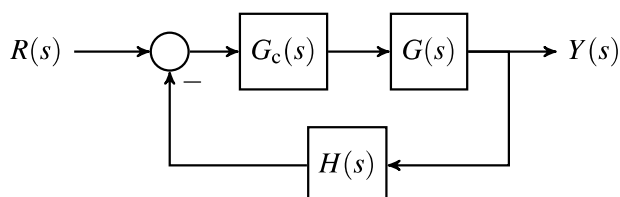
when the transfer function is simplified to a ratio of polynomials the solution is considered an open-loop representation, such as



Both models are valid representations and there's no loss of generality or information with either model, so we may transform between the two forms as needed for analysis and design. The biggest distinction between the two representations is the final form. The closed-loop form retains the forward path transfer function, which is $G_c(s)$ and $G(s)$ cascaded, and the loop-gain transfer function, which is $G_c(s)$, $G(s)$, and $H(s)$ cascaded or $L(s) = G_cGH(s)$. The primary reason we consider the closed-loop form is to analyze the loop-gain. So, if a transfer function is closed-loop the loop-gain is determined by direct observation. While the open-loop transfer function is always $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$ where $B(s)$ and $A(s)$ are polynomials. Additionally, $A(s)$ has a special designation, it is the **characteristic polynomial**.

Example 1

Given:



where

$$G_c(s) = \frac{3}{s+3}$$

$$G(s) = \frac{10}{s+1}$$

and

$$H(s) = \frac{2}{s+2}$$

Find: the open-loop transformation

$$G_c G(s) = \frac{3 \cdot 10}{(s+3)(s+1)} = \frac{30}{s^2 + 4s + 3}$$

and

$$G_c GH(s) = \frac{3 \cdot 10 \cdot 2}{(s+3)(s+1)(s+2)} = \frac{60}{s^3 + 6s^2 + 11s + 6}$$

The single block diagram solution, which is $T(s) = \frac{G_c G(s)}{1 + G_c GH(s)}$ is

$$T(s) = \frac{\frac{30}{s^2 + 4s + 3}}{1 + \frac{60}{s^3 + 6s^2 + 11s + 6}} \quad (3.1)$$

which may be simplified

$$T(s) = \frac{\frac{30}{s^2 + 4s + 3}}{\frac{s^3 + 6s^2 + 11s + 6 + 60}{s^3 + 6s^2 + 11s + 6}} = \frac{30(s+2)}{s^3 + 6s^2 + 11s + 66}$$

written in polynomial form as

$$T(s) = \frac{B(s)}{A(s)} = \frac{30s + 60}{s^3 + 6s^2 + 11s + 66} \quad (3.2)$$

Now, we have two solutions (3.1) and (3.2). The first solution (3.1) still has a distinct representation for the forward path $G_c G(s)$ and the loop-gain

$$L(s) = G_c GH(s)$$

which means it is in closed-loop form. The loop-gain is

$$L(s) = \frac{60}{s^3 + 6s^2 + 11s + 6}$$

The second solution (3.2) has lost the individual representations for the transfer functions, so it is considered the open-loop solution – the open-loop solution is always a ratio of polynomials.

Example 2

Given a transfer function in open-loop form:

$$\frac{Y(s)}{R(s)} = T(s) = \frac{s+4}{s^2+s+9}$$

Find: the closed-loop transformation

Use the **continued fraction model** (Section 2.1) to determine the closed-loop model.

We begin by dividing the numerator and denominator in the open-loop transfer function by the numerator

$$T(s) = \frac{1}{\frac{s^2+s+9}{s+4}}$$

then separate the denominator fraction into two expressions. Take care to make one expression as simple as possible – a constant if possible, and the lowest order

$$T(s) = \frac{1}{\frac{s^2+s}{s+4} + \frac{9}{s+4}}$$

now set it equal to the continued fraction model of a negative feedback block diagram

$$T(s) = \frac{1}{\frac{s^2+s}{s+4} + \frac{9}{s+4}} = \frac{1}{\frac{1}{G(s)} + H(s)}$$

Select the simplest expression for the sensor transfer function and the other as the forward path transfer function $G(s)$

$$H(s) = \frac{9}{s+4}$$

and

$$G(s) = \frac{s+4}{s^2+s}$$

which produces a loop-gain

$$L(s) = GH(s) = \left(\frac{s+4}{s^2+s} \right) \left(\frac{9}{s+4} \right) = \frac{9}{s^2+s}$$

The closed-loop representation is then

$$\frac{Y(s)}{R(s)} = \frac{\frac{s+4}{s^2+s}}{1 + \frac{9}{s^2+s}}$$

Example 3

Given a transfer function in open-loop form:

$$\frac{Y(s)}{R(s)} = T(s) = \frac{4}{s+13}$$

Find: the closed-loop transformation

Just as before use the **continued fraction model** (Section 2.1) to determine the loop-gain.

$$T(s) = \frac{1}{\frac{s+13}{4}}$$

then separate the denominator fraction into two expressions.

$$T(s) = \frac{1}{\frac{s}{4} + \frac{13}{4}}$$

now set it equal to the continued fraction model of a negative feedback block diagram

Select the simplest expression for the sensor transfer function and the other as the forward path transfer function $G(s)$

$$H(s) = \frac{13}{4}$$

and

$$G(s) = \frac{4}{s}$$

which produces a loop-gain

$$L(s) = GH(s) = \frac{13}{s}$$

and a closed-loop transfer function

$$\frac{Y(s)}{R(s)} = \frac{\frac{4}{s}}{1 + \frac{13}{s}}$$

Example 4 — Biproper

Given a transfer function in open-loop form:

$$\frac{Y(s)}{R(s)} = T(s) = \frac{s^2 + 3s + 9}{s^2 - 2s - 7}$$

Find: the closed-loop transformation

The solution to this problem is not as easy as the previous problems because the open-loop transfer function is **biproper** – the order of the numerator is the same as the order of the denominator. Now, our first step is perform long division on transfer function

$$T(s) = \frac{s^2 + 3s + 9}{s^2 - 2s - 7} = 1 + \frac{5s + 16}{s^2 - 2s - 7} = 1 + \hat{T}(s)$$

Now, we have two expressions that represent $T(s)$ and we can use the **continued fraction** model to determine the loop-gain of $\hat{T}(s)$.

$$\hat{T}(s) = \frac{1}{\frac{s^2 - 2s - 7}{5s + 16}}$$

then separate the denominator fraction into two expressions.

$$\hat{T}(s) = \frac{1}{\frac{s^2 - 2s}{5s + 16} - \frac{7}{5s + 16}}$$

Select the simplest expression for the sensor transfer function and the other as the forward path transfer function $G(s)$

$$H(s) = -\frac{7}{5s + 16}$$

and

$$G(s) = \frac{5s + 16}{s^2 - 2s}$$

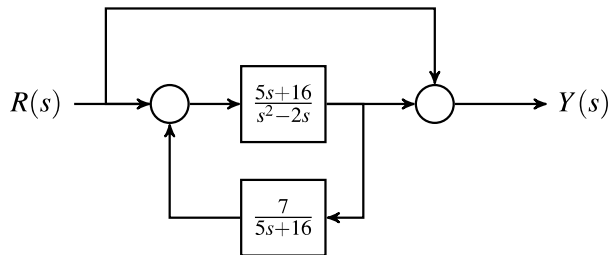
which produces a loop-gain

$$L(s) = GH(s) = -\frac{7}{s^2 - 2s}$$

and a closed-loop transformation

$$\frac{Y(s)}{R(s)} = 1 + \frac{\frac{5s + 16}{s^2 - 2s}}{1 - \frac{7}{s^2 - 2s}}$$

Let's take this example a step further by developing the block diagram. We have $G(s)$ and $H(s)$ defined for the partial expression $\hat{T}(s)$, but must also consider the constant in $T(s)$, which is a path directly from the input signal to the output signal.



Note, the feedback through $H(s)$ is positive due to the negative sign in the $H(s)$ transfer function.

Example 5 — Improper

Given a transfer function in open-loop form:

$$\frac{Y(s)}{R(s)} = T(s) = \frac{s^3 + s^2 + 9s + 9}{s^2 + 4s + 2}$$

Find: the closed-loop transformation

The denominator is higher order than the numerator, which means the open-loop transfer function is **improper**. There is no way to develop a closed-loop model for an improper open-loop transfer function.

3.1 Closed-Loop Models

When using the continued fraction model to transform from open-loop to closed-loop you are presented with a choice for $G(s)$ and $H(s)$. The choice should always be simplest expression for sensor feedback $H(s)$. Ideally, the sensor feedback should be a static gain of one — a.k.a. unity feedback. If unity feedback is not possible due to the system model, then select the simplest model possible — usually a constant over a low-order polynomial. Choosing the simplest model for $H(s)$ does two things:

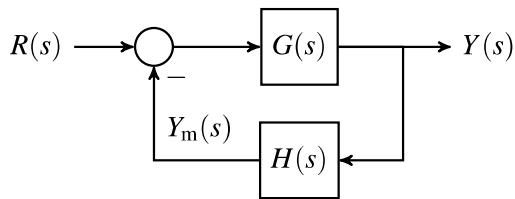
1. Minimizes the feedback circuits
 - The simpler the feedback the less lag is introduced
2. Avoids differentiation
 - Amplifies high frequency noise

Differentiation and Noise Example

Given an open-loop transfer function:

$$\frac{Y(s)}{R(s)} = \frac{1}{s + 13}$$

Find: the closed-loop block diagram



Using the continued fraction model approach we have two options

$$\frac{1}{s + 13} = \frac{1}{\frac{1}{G(s)} + H(s)}$$

Option 1

$$G(s) = \frac{1}{s}$$
$$H(s) = 13$$

which produces an integrator in the forward path.

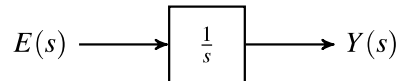
Option 2

$$G(s) = \frac{1}{13}$$
$$H(s) = s$$

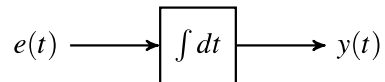
produces a differentiator in the feedback path.

Now, the mathematics tells us the two options are equivalent. However, **Option 1** has the simpler solution for $H(s)$

Let's consider the forward path of **Option 1**



and in the time-domain



where the input error signal is corrupted by noise

$$e = \text{signal} + \text{noise}$$

or

$$e(t) = \sin(t) + 0.01 \sin(10^{10}t)$$

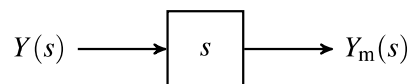
So the output

$$y(t) = \int_0^t e(t) dt = \int_0^t \sin(t) + 0.01 \sin(10^{10}t) dt = \int_0^t \sin(t) dt + \int_0^t 0.01 \sin(10^{10}t) dt$$

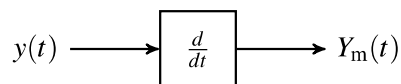
which is

$$y(t) = \text{signal} + \text{noise} = -\cos(t) - \frac{0.01}{10^{10}} \cos(10^{10}t) \approx \text{signal}$$

Now, let's consider the feedback path of **Option 2**



and in the time-domain



where the output signal is corrupted by noise

$$y = \text{signal} + \text{noise}$$

or

$$y(t) = \sin(t) + 0.01 \sin(10^{10}t)$$

So the output

$$Y_m(t) = \frac{dy}{dt} = \frac{d}{dt} \sin(t) + 0.01 \sin(10^{10}t)dt = \frac{d}{dt} \sin(t)dt + \frac{d}{dt} 0.01 \sin(10^{10}t)dt$$

which is

$$Y_m(t) = \text{signal} + \text{noise} = \cos(t) + (0.01 \cdot 10^{10}) \cos(10^{10}t) \approx \text{noise}$$

What we find is that the integrator in the presence of high-frequency noise produces a signal that is fully corrupted by the noise. In electronics, high-frequency noise is common. Switching power supplies generally have a 200 kHz to 2 MHz clock. Additionally, oscillators, time-sync circuits, and timers produce high-frequency noise as well.

3.1.1 Minimal Realization

Block diagram manipulation produce an open-loop transfer function where the zeros and poles will cancel. Always be careful in taking this step to reduce the complexity of the model with zero/pole cancelation. Only stable poles may be canceled in the open-loop transfer function. Unstable poles should *never* be canceled.

Stable Cancellation

If

$$\begin{aligned} T(s) &= \frac{s^2 + 7s + 6}{s^3 + 18s^2 + 104s + 192} \\ &= \frac{(s+1)(s+6)}{(s+8)(s+6)(s+4)} \end{aligned}$$

We can simplify to

$$\begin{aligned} T(s) &= \frac{s+6}{(s+8)(s+4)} \\ &= \frac{s+6}{s^2 + 12s + 32} \end{aligned}$$

Unstable Realization

If

$$\begin{aligned} T(s) &= \frac{s^2 - 5s - 6}{s^3 + 6s^2 - 40s - 192} \\ &= \frac{(s+1)(s-6)}{(s+8)(s-6)(s+4)} \end{aligned}$$

We adjust the model to

$$T(s) = \frac{(s+1)(s-6+\epsilon)}{(s+8)(s-6+\epsilon)(s+4)}$$